

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279366

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

KANG; Myungsam et al.

METHOD OF FABRICATING A SEMICONDUCTOR PACKAGE

Abstract

A semiconductor package and associated methods, the package including a substrate; first and second semiconductor chips on the substrate; and external terminals below the substrate, wherein the substrate includes a core portion; first and second buildup portions on top and bottom surfaces of the core portion, the first and second buildup portions including a dielectric pattern and a line pattern; and an interposer chip in an embedding region in the core portion and electrically connected to the first and second buildup portions, the interposer chip includes a base layer; a redistribution layer on the base layer; and a via that penetrates the base layer, the via being connected to the redistribution layer and exposed at a surface of the base layer, the redistribution layer is connected to a line pattern of the first buildup portion, and the via is connected to a line pattern of the second buildup portion.

Inventors: KANG; Myungsam (Hwaseong-si, KR), KO; Youngchan (Seoul, KR), JEONG; Taesung (Osan-si, KR)

Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Family ID: 80114008

Appl. No.: 19/209877

Filed: May 16, 2025

Foreign Application Priority Data

KR 10-2020-0099253

Aug. 07, 2020

Related U.S. Application Data

parent US continuation 18367039 20230912 parent-grant-document US 12334445 child US 19209877

parent US continuation 17183562 20210224 parent-grant-document US 11784129 child US 18367039

Publication Classification

Int. Cl.: **H01L23/538** (20060101); **H01L21/48** (20060101); **H01L23/00** (20060101); **H01L23/31** (20060101); **H01L23/498** (20060101)

U.S. Cl.:

CPC **H01L23/5383** (20130101); **H01L21/4853** (20130101); **H01L21/4857** (20130101); **H01L21/486** (20130101); **H01L23/3107** (20130101); **H01L23/49822** (20130101); **H01L23/49838** (20130101); **H01L23/5386** (20130101); **H01L24/20** (20130101); **H01L2224/211** (20130101); **H01L2224/214** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/367,039, filed Sep. 12, 2023, which is a continuation of U.S. application Ser. No. 17/183,562, filed Feb. 24, 2021, now U.S. Pat. No. 11,784,129, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0099253, filed Aug. 7, 2020, in the Korean Intellectual Property Office, the entire contents of all of which are hereby incorporated by reference. [0002] Korean Patent Application No. 10-2020-0099253, filed on Aug. 7, 2020, in the Korean Intellectual Property Office, and entitled: “Semiconductor Package and Method of Fabricating the Same,” is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0003] Embodiments relate to a semiconductor package and a method of fabricating the same.

2. Description of the Related Art

[0004] With the development of the electronics industry, electronic products have increasingly exhibited high performance, high speed, and compact size. To meet the trend, a packaging technology in which a plurality of semiconductor chips are mounted in a single package has been considered.

[0005] A semiconductor package may be provided to implement an integrated circuit chip to qualify for use in electronic products.

SUMMARY

[0006] The embodiments may be realized by providing a semiconductor package including a substrate; a first semiconductor chip and a second semiconductor chip that are mounted on the substrate; and a plurality of external terminals below the substrate, wherein the substrate includes a core portion; a first buildup portion and a second buildup portion respectively on top and bottom surfaces of the core portion, each of the first and second buildup portions including a dielectric pattern and a line pattern; and an interposer chip in an embedding region in the core portion and electrically connected to the first and second buildup portions, wherein the interposer chip includes a base layer; a redistribution layer on a first surface of the base layer; and a via that penetrates the base layer, the via being connected to the redistribution layer and exposed at a second surface of the base layer, wherein the redistribution layer is connected to the line pattern of the first buildup portion, and wherein the via is connected to the line pattern of the second buildup portion.

[0007] The embodiments may be realized by providing a semiconductor package including a first wiring layer; a second wiring layer on the first wiring layer; an interposer chip between the first wiring layer and the second wiring layer; an insulating layer between the first wiring layer and the second wiring layer, the insulating layer surrounding the interposer chip; a conductive element

between the first wiring layer and the second wiring layer, the conductive element being spaced apart from the interposer chip and connecting the first wiring layer to the second wiring layer; a first semiconductor chip and a second semiconductor chip that are mounted on the second wiring layer; a molding layer on the second wiring layer, the molding layer covering the first and second semiconductor chips; and a plurality of external terminals below the first wiring layer, wherein the interposer chip includes a base layer; a redistribution layer on the base layer and coupled to the second wiring layer; and a via in the base layer, the via connecting the redistribution layer to the first wiring layer, and wherein the first semiconductor chip and the second semiconductor chip are electrically connected to each other through the interposer chip.

[0008] The embodiments may be realized by providing a method of fabricating a semiconductor package, the method including forming an interposer chip that includes a base layer, a redistribution layer on a first surface of the base layer, and a via that penetrates the base layer, the via being connected to the redistribution layer and exposed on a second surface of the base layer; providing the interposer chip into a core portion that includes a conductive element; forming a first buildup portion on a top surface of the core portion and the first surface of the base layer, the first buildup portion being connected to the conductive element and the redistribution layer; forming a second buildup portion on a bottom surface of the core portion and the second surface of the base layer, the second buildup portion being connected to the conductive element and the via; mounting a plurality of semiconductor chips on the first buildup portion; and forming on the first buildup portion a molding layer that covers the plurality of semiconductor chips, wherein the plurality of semiconductor chips are electrically connected to each other through the first buildup portion and the interposer chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Features will be apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:

[0010] FIG. 1 is a cross-sectional view of a semiconductor package according to some example embodiments.

[0011] FIG. 2 is an enlarged view showing section A of FIG. 1.

[0012] FIGS. 3 to 5 are cross-sectional views of a semiconductor package according to some example embodiments.

[0013] FIGS. 6 to 11 are cross-sectional views of stages in a method of fabricating an interposer chip according to some example embodiments.

[0014] FIGS. 12 to 17 are cross-sectional views of stages in a method of fabricating a semiconductor package according to some example embodiments.

[0015] FIGS. 18 to 21 are cross-sectional views of stages in a method of fabricating a semiconductor package according to some example embodiments.

DETAILED DESCRIPTION

[0016] FIG. 1 is a cross-sectional view of a semiconductor package according to some example embodiments. FIG. 2 is an enlarged view showing section A of FIG. 1. FIG. 3 is a cross-sectional view of a semiconductor package according to some example embodiments.

[0017] Referring to FIGS. 1 and 2, a semiconductor package 1 may include a wiring substrate 100, first and second semiconductor chips 200 and 300 mounted on the wiring substrate 100, and a molding layer 400 that covers the first and second semiconductor chips 200 and 300 on the wiring substrate 100.

[0018] The wiring substrate 100 may be provided. The wiring substrate 100 may include a core portion 110, an interposer chip 120 in the core portion 110, an upper buildup portion 130 on one

surface (e.g., a top surface) of the core portion **110**, and a lower buildup portion **140** on another surface (e.g., a bottom surface) of the core portion **110**.

[0019] The core portion **110** may extend (e.g., lengthwise) in one direction. The core portion **110** may include a core pattern whose part is removed when viewed in plan. A region CA from which the part is removed may correspond to an embedding region CA where the interposer chip **120** is disposed as discussed below. The embedding region CA may be shaped like an open hole that connects the top and bottom surfaces of the core portion **110** to each other (e.g., that extends completely through the core portion **110**). In an implementation, the core portion **110** may include one core pattern. In an implementation, the core portion **110** may include two or more core patterns. In an implementation, the wiring substrate **100** may include a plurality of core patterns that are spaced apart from each other when viewed in plan. The core portion **110** may include a dielectric material. In an implementation, the core portion **110** may include glass fibers, ceramic plates, epoxy, or resins. In an implementation, the core portion **110** may include stainless steel, aluminum (Al), nickel (Ni), magnesium (Mg), zinc (Zn), tantalum (Ta), or any combination thereof. As used herein, the term “or” is not an exclusive term, e.g., “A or B” would include A, B, or A and B.

[0020] The core portion **110** may include a core dielectric pattern **112** and a core conductive pattern **114** (e.g., a wiring pattern in the core dielectric pattern **112**). In an implementation, the core dielectric pattern **112** may include silicon oxide (e.g., SiO₂). The core conductive pattern **114** may be spaced apart from the embedding region CA. In an implementation, the core portion **110** may have, at its outer side, the core conductive pattern **114** that is outside the embedding region CA. The core conductive pattern **114** may include upper core pads **115**, core wiring lines **116**, and lower core pads **117**. The upper core pads **115** may be on the top surface of the core portion **110**. The upper core pads **115** may be electrically connected to the upper buildup portion **130** which will be discussed below. The lower core pads **117** may be on a lower part of the core portion **110**. The lower core pads **117** may be electrically connected to the lower buildup portion **140** which will be discussed below. The upper and lower core pads **115** and **117** may have a thickness of about 10 μm to about 20 μm. The core wiring lines **116** may electrically connect the upper core pads **115** to the lower core pads **117**. A core protection layer **119** may further be provided on a bottom surface of the core dielectric pattern **112** of the core portion **110**, thereby covering the lower core pads **117**. In an implementation, the core protection layer **119** may be omitted. In an implementation, the lower core pads **117** may be exposed on the bottom surface of the core portion **110**.

[0021] In an implementation, as illustrated in FIG. 1, the core conductive pattern **114** may be formed buried in a plurality of stacked core dielectric patterns **112**. In an implementation, as shown in FIG. 3, a core portion **110'** may include a single-layered core dielectric pattern **112**, and instead of the core conductive pattern **114** illustrated in FIG. 1, through electrodes **114-1** may vertically penetrate the core dielectric pattern **112**. The through electrodes **114-1** may extend from top to bottom surfaces of the core dielectric pattern **112**. Upper core pads **115** may be coupled to top surfaces of the through electrodes **114-1**, and lower core pads **117** may be coupled to bottom surfaces of the through electrodes **114-1**. The following description will focus on the embodiment shown in FIGS. 1 and 2.

[0022] The interposer chip **120** may be in the embedding region CA of the core portion **110**. The core portion **110** may include the embedding region CA at its part from which the core pattern is removed. The embedding region CA may be exposed on or at the top and bottom surfaces of the core portion **110**. In an implementation, the embedding region CA may be shaped like a through hole that extends from the top surface of the core portion **110** toward the bottom surface of the core portion **110**. In an implementation, the embedding region CA may have a recessed shape that extends toward an inside of the core portion **110**. In this case, the embedding region CA may have a floor surface that corresponds to the core protection layer **119**.

[0023] The interposer chip **120** may include a base layer **121**, a redistribution layer **123** on the base

layer **121**, and vias **126** in the base layer **121**.

[0024] The base layer **121** may include silicon (Si). The base layer **121** may have a thickness of about 70 μm to about 300 μm . A chip protection layer **122** may cover a bottom surface **121b** of the base layer **121**. The chip protection layer **122** may be a partial section of a lower part of the base layer **121**. In an implementation, the chip protection layer **122** may be a component that is formed when the lower part of the base layer **121** is partially oxidized.

[0025] The interposer chip **120** may have a top surface and a bottom surface. In this description below, the language “top surface” may be defined to refer to a surface on which are formed wiring line patterns of the interposer chip **120**, and the language “bottom surface” may be defined to refer to an opposing surface that faces the top surface. In an implementation, the interposer chip **120** may include the redistribution layer **123** on or at the top surface thereof. The redistribution layer **123** may be on a top surface **121a** of the base layer **121**. The redistribution layer **123** may include a chip conductive pattern **124** on the top surface of the interposer chip **120**, and may also include a chip dielectric layer **125** that covers the chip conductive pattern **124**. The chip conductive pattern **124** may have a part that is exposed on a top surface of the chip dielectric layer **125**, and the exposed part of the chip conductive pattern **124** may correspond to upper chip pads **124a**, **124b**, and **124c**. The upper chip pads **124a**, **124b**, and **124c** may have their exposed surfaces coplanar with that of the chip dielectric layer **125**. The chip conductive pattern **124** may include a conductive material, e.g., a metal. In an implementation, the chip conductive pattern **124** may include copper (Cu). In an implementation, the chip dielectric layer **125** may include an oxide. In an implementation, the chip dielectric layer **125** may include silicon oxide (SiOx).

[0026] The vias **126** may vertically penetrate the base layer **121**. The vias **126** may have their ends (or bottom surfaces) **126b** that are exposed on or at the bottom surface of the interposer chip **120**. The bottom surfaces **126b** of the vias **126** may be coplanar with the bottom surface of the interposer chip **120** (or, with the bottom surface **121b** of the chip protection layer **122**, e.g., which was the bottom surface of the base layer **121** prior to forming the protection layer **122**), and the bottom surfaces **126b** of the vias **126** and the bottom surface of the interposer chip **120** may be substantially flat. The vias **126** may have their other ends (or top surfaces) **126a** that extend toward the top surface of the interposer chip **120** and contact the redistribution layer **123**. The top surfaces **126a** of the vias **126** may be located at the same level as that of (e.g., may be coplanar with) the top surface **121a** of the base layer **121**, may be in contact with a bottom surface of the redistribution layer **123**, and may be located at the same level as that of the bottom surface of the redistribution layer **123**. In an implementation, the redistribution layer **123** may be higher than the vias **126**. The vias **126** may be coupled to the chip conductive pattern **124** of the redistribution layer **123**. The vias **126** may have a width of about 50 μm to about 150 μm .

[0027] The interposer chip **120** may further include lower chip pads **127** on or at the bottom surface thereof. The lower chip pads **127** may be on the bottom surface **121b** of the chip protection layer **122**. The lower chip pads **127** may be aligned with the vias **126**, and the vias **126** may be connected to the lower chip pads **127** to the chip conductive pattern **124** of the redistribution layer **123**. A seed layer **128** may be on each of top surfaces of the lower chip pads **127**. The seed layers **128** may be between the lower chip pads **127** and the vias **126** and between the lower chip pads **127** and the chip protection layer **122**.

[0028] The interposer chip **120** may be spaced apart at a certain distance from an inner wall of the embedding region CA of the core portion **110**, and may be surrounded by the inner wall of the embedding region CA of the core portion **110**. In an implementation, when viewed in plan, the core portion **110** may have a shape that surrounds the interposer chip **120**. The interposer chip **120** may be provided in a face-up manner. In an implementation, the interposer chip **120** may be disposed to allow the redistribution layer **123** to face the top surface of the core portion **110**. The interposer chip **120** may have a thickness less than that of the core portion **110**. In an implementation, the interposer chip **120** may have a thickness the same as or less than that of the core portion **110**. The

upper chip pads **124a**, **124b**, and **124c** of the interposer chip **120** may have top surfaces at a higher level than that of a top surface of the core dielectric pattern **112** of the core portion **110**. The upper chip pads **124a**, **124b**, and **124c** of the interposer chip **120** may be located at the same level as that of the upper core pads **115** of the core portion **110**. The bottom surface of the interposer chip **120** may be located at a higher level than that of the bottom surface of the core portion **110**. In an implementation, the lower chip pads **127** of the interposer chip **120** may have their bottom surfaces at a higher level than that of a bottom surface of the core protection layer **119**.

[0029] In an implementation, when the embedding region CA has a recessed shape that extends toward an inside of the core portion **110**, the interposer chip **120** may be placed on the floor surface of the embedding region CA. In this case, an adhesive agent or an adhesive film may be used to attach the interposer chip **120** to the floor surface of the embedding region CA.

[0030] The interposer chip **120** may not include an integrated circuit. In an implementation, the interposer chip **120** may be a component that facilitates wiring of the wiring substrate **100**. A semiconductor integration circuit (IC) process may be used to form the redistribution layer **123** of the interposer chip **120** including silicon (Si), and thus the redistribution layer **123** of the interposer chip **120** may have a high line density. Therefore, the wiring substrate **100** may have a high degree of freedom of wiring. In addition, highly thermal conductive silicon (Si) and the vertically extending vias **126** may facilitate a downward discharge of heat generated from the first and second semiconductor chips **200** and **300** mounted on the wiring substrate **100**. As a result, the semiconductor package **1** may have high thermal radiation efficiency.

[0031] In an implementation, the interposer chip **120** may include a certain integrated circuit. In this case, the integrated circuit may be on the top surface of the interposer chip **120**. The integrated circuit may include a ground circuit or a power circuit. In an implementation, the integrated circuit may include a memory circuit, a logic circuit, or a radio frequency integrated circuit.

[0032] The embedding region CA may have therein an insulating layer **113** that fills a space between the core portion **110** and the interposer chip **120**. The insulating layer **113** may extend below the interposer chip **120** and the core portion **110**. In an implementation, the insulating layer **113** may cover the bottom surface of the interposer chip **120** and the bottom surface of the core portion **110**. In an implementation, the insulating layer **113** may expose (e.g., may not cover or contact) the bottom surface of the interposer chip **120** and the bottom surface of the core portion **110**. The insulating layer **113** may include an insulating polymer. In an implementation, the insulating layer **113** may include a thermosetting resin such as an epoxy resin, a thermoplastic resin such as polyimide, or a resin, such as Ajinomoto buildup film (ABF), flame retardant-4 (FR-4), or bismaleimide triazine (BT), including a stiffener such as an inorganic filler. In an implementation, the insulating layer **113** may include a molding material such as epoxy molding compound (EMC) or a photosensitive material such as PIE (photo-imageable epoxy).

[0033] The upper and lower buildup portions **130** and **140** may respectively cover the top and bottom surfaces of the core portion **110**. The upper buildup portion **130** may cover the top surface of the core portion **110** and the top surface of the interposer chip **120**. The lower buildup portion **140** may cover the bottom surface of the core portion **110** and the bottom surface of the interposer chip **120**.

[0034] The upper buildup portion **130** may include upper dielectric patterns **132** and upper line patterns **134** that are sequentially stacked on the top surface of the core portion **110**. The upper buildup portion **130** may cover the top surface of the core portion **110** and the redistribution layer **123** of the interposer chip **120**. The lower buildup portion **140** may include lower dielectric patterns **142** and lower line patterns **144** that are sequentially stacked on the bottom surface of the core portion **110**. The lower buildup portion **140** may cover the bottom surface of the core portion **110** and the chip dielectric layer **125** of the interposer chip **120**. The upper and lower dielectric patterns **132** and **142** may include prepreg, Ajinomoto build-up films (ABF), flame retardant-4 (FR-4), or bismaleimide triazine (BT). The upper and lower line patterns **134** and **144** may include a circuit

pattern. The upper line patterns **134** may be connected to the upper core pads **115** of the core portion **110** and to the upper chip pads **124a**, **124b**, and **124c** of the interposer chip **120**. In an implementation, the upper buildup portion **130** may contact the top surface of the core portion **110** and the top surface of the interposer chip **120**, and the upper line patterns **134** may contact the upper core pads **115** and the upper chip pads **124a**, **124b**, and **124c**. The lower line patterns **144** may be connected to the lower core pads **117** of the core portion **110** and to the lower chip pads **127** of the interposer chip **120**. In an implementation, the lower buildup portion **140** may contact a bottom surface of the insulating layer **113**, and the lower line patterns **144** may penetrate the insulating layer **113** to come into contact with the lower chip pads **127** and the lower core pads **117**. The upper and lower line patterns **134** and **144** may include copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or any combination thereof.

[0035] External terminals **150** may be below the lower buildup portion **140**. The external terminals **150** may be on a bottom surface of the lower buildup portion **140**. In an implementation, the external terminals **150** may penetrate the lower dielectric patterns **142** of the lower buildup portion **140** and may be coupled to the lower line patterns **144**. In an implementation, the external terminals **150** may be on substrate pads on the bottom surface of the lower buildup portion **140**. In this case, the substrate pads may be either ones of the lower line patterns **144** exposed from the lower dielectric patterns **142** of the lower buildup portion **140** or separate pads that are on the lower dielectric patterns **142** of the lower buildup portion **140** and are connected to the lower line patterns **144**. The external terminals **150** may be electrically connected through the lower line patterns **144** to the interposer chip **120** and the core portion **110**. The external terminals **150** may include a solder ball or a solder bump.

[0036] The first semiconductor chip **200** and the second semiconductor chip **300** may be mounted on the wiring substrate **100**. The first semiconductor chip **200** and the second semiconductor chip **300** may be spaced apart from each other. The first semiconductor chip **200** and the second semiconductor chip **300** may include a semiconductor material, e.g., silicon (Si). The first semiconductor chip **200** may include a first circuit layer **210** and the second semiconductor chip **300** may include a second circuit layer **310**. The first circuit layer **210** of the first semiconductor chip **200** may include a logic circuit. In an implementation, the first semiconductor chip **200** may be a logic chip. The second circuit layer **310** of the second semiconductor chip **300** may include a memory circuit. In an implementation, the second semiconductor chip **300** may be a memory chip. The second semiconductor chip **300** may be provided in plural. In an implementation, the plurality of second semiconductor chips **300** may have a vertical chip stack structure. The first semiconductor chip **200** and the second semiconductor chip **300** may be flip-chip mounted on the wiring substrate **100**. In an implementation, first chip terminals **230** may be on first chip pads **220** on one surface of the first circuit layer **210** of the first semiconductor chip **200**. Second chip terminals **330** may be on second chip pads **320** on one surface of the second circuit layer **310** of the second semiconductor chip **300**. The first chip terminals **230** and the second chip terminals **330** may be coupled to the upper line patterns **134** of the upper buildup portion **130**. In an implementation, ones of the upper line patterns **134** may be pads exposed on a top surface of the upper dielectric patterns **132**, and the first chip terminals **230** and the second chip terminals **330** may connect the exposed upper line patterns **134** to the first and second chip pads **220** and **320**. In this case, the first semiconductor chip **200** may be connected through the interposer chip **120** to the second semiconductor chip **300**. In an implementation, the first semiconductor chip **200** may be coupled through the upper buildup portion **130** to the first upper chip pad **124a** of the redistribution layer **123** included in the interposer chip **120**. The second semiconductor chip **300** may be coupled through the upper buildup portion **130** to the second upper chip pad **124b** of the redistribution layer **123** included in the interposer chip **120**. The first upper chip pad **124a** and the second upper chip pad **124b** may be electrically connected to each other in the redistribution layer **123**. Hence, a chip-to-chip connection may be achieved between the first semiconductor chip **200** and the second

semiconductor chip **300**. As the first semiconductor chip **200** and the second semiconductor chip **300** are connected to each other through the redistribution layer **123** having a high line density in the interposer chip **120**, a reduced electrical connection length may be accomplished between the first semiconductor chip **200** and the second semiconductor chip **300**, and the semiconductor package **1** may have improved electrical characteristics.

[0037] The first and second upper chip pads **124a** and **124b** that connect the first and second semiconductor chips **200** and **300** to each other may be electrically floated from the vias **126** of the interposer chip **120**. The third upper chip pads **124c** (other than the first and second upper chip pads **124a** and **124b** among the upper chip pads **124a**, **124b**, and **124c**) may be electrically connected to the vias **126**.

[0038] In an implementation, the first and second upper chip pads **124a** and **124b** that connect the first and second semiconductor chips **200** and **300** to each other may be electrically connected to the vias **126** of the interposer chip **120**. In an implementation, the first semiconductor chip **200** and the second semiconductor chip **300** may receive a power or ground signal that is externally applied through the lower buildup portion **140**, the vias **126**, and the third upper chip pads **124c**. The power or ground signal may be transferred to the first and second semiconductor chips **200** and **300** through the interposer chip **120** that is individually separated from the core portion **110** that serves as an electrical path along which a drive signal is transferred. Accordingly, it is possible to easily group a power or ground circuit for equalization of the power or ground signal transferred to the first and second semiconductor chips **200** and **300**, to effortlessly achieve a wiring design for the drive signal or for the power and ground signals, and to increase the degree of freedom of wiring.

[0039] The molding layer **400** may be on the wiring substrate **100**. The wiring substrate **100** may be provided on its top surface with the molding layer **400** that covers the first and second semiconductor chips **200** and **300**. The molding layer **400** may include an insulating material. In an implementation, the molding layer **400** may include an epoxy molding compound (EMC).

[0040] The semiconductor package **1** may be provided as discussed above.

[0041] FIG. **4** is a cross-sectional view of a semiconductor package according to some example embodiments. In the embodiments that follow, a repeated detailed description of technical features discussed above with reference to FIGS. **1** to **3** may be omitted, and a difference thereof may be discussed in detail. The same components as those of the semiconductor package discussed above will be allocated the same reference numerals thereto.

[0042] Referring to FIG. **4**, a semiconductor package **2** may include a wiring substrate **100**, first and second semiconductor chips **200** and **300** mounted on the wiring substrate **100**, and a molding layer **400** that covers the first and second semiconductor chips **200** and **300** on the wiring substrate **100**.

[0043] The wiring substrate **100** may be provided. The wiring substrate **100** may include an interposer chip **120**, an upper buildup portion **130** on the interposer chip **120**, and a lower buildup portion **140** below the interposer chip **120**.

[0044] The interposer chip **120**, the upper buildup portion **130**, and the lower buildup portion **140** may be substantially the same as or similar to those discussed with reference to FIGS. **1** to **3**.

[0045] In an implementation, the interposer chip **120** may include the base layer **121**, the redistribution layer **123** on the base layer **121**, and the vias **126** in the base layer **121**. The base layer **121** may include silicon (Si). The redistribution layer **123** may be on the top surface **121a** of the base layer **121**. The vias **126** may penetrate the base layers **121** and may be coupled to the redistribution layer **123**. The lower chip pads **127** may be on the bottom surface **121b** of the chip protection layer **122** and may be aligned with the vias **126**.

[0046] The upper buildup portion **130** may include the upper dielectric patterns **132** and the upper line patterns **134** that are sequentially stacked on the interposer chip **120**. The upper line patterns **134** may be connected to the upper chip pads **124a**, **124b**, and **124c** of the interposer chip **120**.

[0047] The lower buildup portion **140** may include the lower dielectric patterns **142** and the lower

line patterns **144** that are sequentially stacked below the interposer chip **120**. The lower line patterns **144** may be connected to the lower chip pads **127** of the interposer chip **120**.

[0048] An insulating layer **113-1** may fill a space between the lower buildup portion **140** and the upper buildup portion **130**. The insulating layer **113-1** may bury the interposer chip **120** between the lower buildup portion **140** and the upper buildup portion **130**. In an implementation, when viewed in plan, the insulating layer **113-1** may surround the interposer chip **120**. The insulating layer **113-1** may contact a lateral surface of the interposer chip **120**. A part of the insulating layer **113-1** may cover the bottom surface of the interposer chip **120**. In an implementation, the lower chip pads **127** of the interposer chip **120** may not be exposed. The insulating layer **113-1** may include an insulating material, e.g., an epoxy molding compound (EMC). In an implementation, the insulating layer **113-1** may include an insulating polymer. In an implementation, the insulating layer **113-1** may include a thermosetting resin such as an epoxy resin, a thermoplastic resin such as polyimide, or a resin, such as Ajinomoto buildup film (ABF), flame retardant-4 (FR-4), or bismaleimide triazine (BT), including a stiffener such as an inorganic filler. In an implementation, the insulating layer **113-1** may include a molding material such as an epoxy molding compound (EMC) or a photosensitive material such as photo-imageable epoxy (PIE).

[0049] Through electrodes **114-2** may be in the insulating layer **113-1**. At a side of the interposer chip **120**, the through electrodes **114-2** may vertically penetrate the insulating layer **113-1**. The through electrodes **114-2** may have a width that decreases as approaching the upper buildup portion **130** (e.g., the through electrodes **114-2** may have a tapered shape). The through electrodes **114-2** may penetrate the insulating layer **113-1** and may be exposed on or at a top surface of the insulating layer **113-1**. The upper line patterns **134** of the upper buildup portion **130** may penetrate the upper dielectric patterns **132** and may contact top surfaces of the through electrodes **114-2**. The through electrodes **114-2** may penetrate the insulating layer **113-1** and may be exposed on or at a bottom surface of the insulating layer **113-1**. The lower line patterns **144** of the lower buildup portion **140** may penetrate the lower dielectric patterns **142** and may contact bottom surfaces of the through electrodes **114-2**. In such a configuration, the lower line pattern **144** in contact with the through electrode **114-2** may be integrally formed with the through electrode **114-2**. The lower line patterns **144** may also penetrate the insulating layer **113-1** and may be coupled to the lower chip pads **127**.

[0050] FIG. 5 is a cross-sectional view of a semiconductor package according to some example embodiments.

[0051] Referring to FIG. 5, a semiconductor package **3** may include a wiring substrate **100**, first and second semiconductor chips **200** and **300** mounted on the wiring substrate **100**, and a molding layer **400** that covers the first and second semiconductor chips **200** and **300** on the wiring substrate **100**.

[0052] The wiring substrate **100**, the first and second semiconductor chips **200** and **300**, and the molding layer **400** may be substantially the same as or similar to those discussed with reference to FIGS. 1 to 3. For example, the wiring substrate **100** may include the core portion **110**, an interposer chip **120** in the embedding region CA of the core portion **110**, the upper buildup portion **130** on the core portion **110**, and the lower buildup portion **140** below the core portion **110**. The first and second semiconductor chips **200** and **300** may be mounted on the upper buildup portion **130** of the wiring substrate **100**. The first and second semiconductor chips **200** and **300** may be connected to each other through the upper buildup portion **130** and the redistribution layer **123** of the interposer chip **120**. On the wiring substrate **100**, the molding layer **400** may cover the first and second semiconductor chips **200** and **300**.

[0053] The semiconductor package **3** may further include an upper package **500** on the molding layer **400**. In an implementation, the semiconductor package **3** may be provided in the structure of package-on-package (POP).

[0054] The upper package **500** may include an upper substrate **510**, an upper semiconductor chip **520**, and an upper molding layer **530**.

[0055] The upper substrate **510** may be a printed circuit board (PCB). In an implementation, the upper substrate **510** may be a redistribution layer. The upper substrate **510** may include first upper substrate pads **512** on a bottom surface of the upper substrate **510**, and may also include second upper substrate pads **514** on a top surface of the upper substrate **510**.

[0056] The upper semiconductor chip **520** may be mounted on the upper substrate **510**. In an implementation, the upper semiconductor chip **520** may be flip-chip mounted on the upper substrate **510**. The upper semiconductor chip **520** may be mounted on the upper substrate **510** through third chip terminals **524** between the second upper substrate pads **514** and upper chip pads **522**. The upper semiconductor chip **520** may include a memory chip or a logic chip. In an implementation, as illustrated in FIG. 5, a single upper semiconductor chip **520** may be provided. In an implementation, the upper semiconductor chip **520** may be provided in plural. In this case, the plurality of upper semiconductor chips **520** may be provided either stacked on one another or horizontally spaced apart from each other on the upper substrate **510**.

[0057] The upper molding layer **530** may be on the upper substrate **510**. On the upper substrate **510**, the upper molding layer **530** may cover the upper semiconductor chip **520**. The upper molding layer **530** may include an insulating material, e.g., an epoxy molding compound (EMC).

[0058] The upper package **500** may be on the wiring substrate **100**. In an implementation, connection terminals **410** may be in the molding layer **400**. The connection terminals **410** may penetrate the molding layer **400** and may connect the first upper substrate pads **512** of the upper substrate **510** to the upper line patterns **134** of the upper buildup portion **130** in the wiring substrate **100**. In an implementation, as illustrated in FIG. 5, the connection terminals **410** may be conductive posts that are formed to vertically penetrate the molding layer **400**. In an implementation, the connection terminals **410** may include a soldering member such as a solder ball or a solder bump, or may have various structures capable of mechanically connecting the first upper substrate pads **512** to the upper line patterns **134**. The upper package **500** may be electrically connected to the external terminals **150** through the connection terminals **410** and the wiring substrate **100**.

[0059] FIGS. 6 to 17 are cross-sectional views of stages in a method of fabricating a semiconductor package according to some example embodiments. Of the FIGS. 6 to 17, FIGS. 6 to 11 are cross-sectional views of a method of fabricating an interposer chip according to some example embodiments.

[0060] The semiconductor package **3** may be provided as discussed above.

[0061] Referring to FIG. 6, a base layer **121** may be provided. The base layer **121** may include, e.g., a semiconductor wafer. In an implementation, the base layer **121** may be a silicon wafer. The base layer **121** may have a first surface **121a** and a second surface **121c** opposite to the first surface **121a**.

[0062] Holes **H** may be formed on the base layer **121**. The holes **H** may be formed by performing a process such as laser drilling on the first surface **121a** of the base layer **121**. In an implementation, the holes **H** may be formed by forming a mask pattern on the first surface **121a** of the base layer **121**, and then using the mask pattern as an etching mask to perform an anisotropic etching process. None of the holes **H** may vertically completely penetrate the base layer **121**. In an implementation, the holes **H** may be formed spaced apart from the second surface **121c**.

[0063] Referring to FIG. 7, vias **126** may be formed in the base layer **121**. In an implementation, a conductive material may be coated or deposited on the first surface **121a** of the base layer **121**. The conductive material may fill the holes **H** of the base layer **121**. Afterwards, a part of the conductive material may be removed from the first surface **121a** of the base layer **121**, and the conductive material may remain only in the holes **H**.

[0064] Referring to FIG. 8, a redistribution layer **123** may be formed on the first surface **121a** of the base layer **121**. In an implementation, a dielectric layer, such as silicon oxide, may be formed on the first surface **121a** of the base layer **121**, and then the dielectric layer may be patterned to form a part of a chip dielectric layer **125**. The chip dielectric layer **125** may expose the vias **126**. A

conductive layer may be formed on a top surface of the chip dielectric layer **125**, and then the conductive layer may be patterned to form a chip conductive pattern **124**. The chip conductive pattern **124** may be electrically connected to the vias **126**. As discussed above, the formation and patterning of the dielectric layer and the formation and patterning of the conductive layer may be repeatedly performed to form the redistribution layer **123**.

[0065] Referring to FIG. **9**, a part of the base layer **121** may be removed. In an implementation, the base layer **121** may be thinned to expose the vias **126**. In an implementation, the second surface **121c** of the base layer **121** may undergo a thinning process, such as grinding or chemical vapor polishing (CMP). The thinning process may cause the base layer **121** to have a third surface **121b** coplanar with ends of the vias **126**. In an implementation, the third surface **121b** of the base layer **121** and the ends of the vias **126** may be substantially flat.

[0066] Referring to FIG. **10**, a chip protection layer **122** may be formed on or in the third surface **121b** of the base layer **121**. In an implementation, an oxidation process or a nitridation process may be performed on the exposed third surface **121b** of the base layer **121**. Therefore, the third surface **121b** of the base layer **121** may be partially oxidized or nitrified to form the chip protection layer **122**. The chip protection layer **122** may have a top surface coplanar with the ends of the vias **126**.

[0067] Lower chip pads **127** may be formed on the chip protection layer **122**. In an implementation, a seed layer **128** may be formed on the chip protection layer **122**. The seed layer **128** may cover the top surface of the chip protection layer **122** and the ends (or top surfaces) of the vias **126**. A mask pattern MP may be formed on the seed layer **128**. The mask pattern MP may have openings that overlap the vias **126**. Thereafter, the seed layer **128** may be used as a seed used for a plating process to form the lower chip pads **127** that fill the openings. The lower chip pads **127** may be formed on the vias **126**.

[0068] Referring to FIG. **11**, the mask pattern MP may be removed. After that, the lower chip pads **127** may be used as an etching mask to pattern the seed layer **128**. Therefore, the seed layer **128** may be patterned to remain between the chip protection layer **122** and the lower chip pads **127** and between the vias **126** and the lower chip pads **127**.

[0069] The process mentioned above may form an interposer chip **120**.

[0070] In an implementation, as illustrated in FIGS. **6** to **11**, a single interposer chip **120** may be formed on the base layer **121**. In an implementation, a plurality of interposer chips **120** may be formed on a single base layer **121**, and in this case, after the formation of the interposer chips **120**, the base layer **121** may undergo a sawing process to separate the interposer chips **120** from each other.

[0071] The following will continuously discuss a method of fabricating a semiconductor package.

[0072] Referring to FIG. **12**, a core portion **110** may be provided. The core portion **110** may include a core dielectric pattern **112** and a core conductive pattern **114** formed in the core dielectric pattern **112**. The core conductive pattern **114** may include upper core pads **115**, core wiring lines **116**, and lower core pads **117**. A core protection layer **119** may be formed on a bottom surface of the core dielectric pattern **112**. In an implementation, a dielectric material may be coated on the bottom surface of the core dielectric pattern **112**, thereby forming the core protection layer **119**.

[0073] A first carrier substrate **900** may be attached to a bottom surface of the core portion **110**. The first carrier substrate **900** may be a dielectric substrate including glass or polymer or a conductive substrate including metal. The first carrier substrate **900** may be provided on its top surface with an adhesive member through which the first carrier substrate **900** may be attached to a bottom surface of the core protection layer **119**. In an implementation, the adhesive member may include a glue tape.

[0074] A protection layer **910** may be formed on the core portion **110**.

[0075] Referring to FIG. **13**, an embedding region CA may be formed in the core portion **110**. The embedding region CA may be formed by removing a partial region of the core portion **110** so as to allow the embedding region CA to penetrate the core portion **110**. In an implementation, the

embedding region CA may be formed by performing an etching process such as drilling, laser ablation, or laser cutting. The removed partial region of the core portion **110** may be a space where the interposer chip **120** is to be provided in a subsequent process.

[0076] Referring to FIG. **14**, the first carrier substrate **900** and the protection layer **910** may be removed from a resultant structure of FIG. **13**.

[0077] A second carrier substrate **920** may be attached to the core portion **110**. The second carrier substrate **920** may be an insulating substrate including glass or polymer or a conductive substrate including metal. The second carrier substrate **920** may be provided on its top surface with an adhesive member through which the second carrier substrate **920** may be attached to the core portion **110**. In an implementation, the adhesive member may include a glue tape. The core portion **110** may be attached to the second carrier substrate **920** so as to allow the upper core pads **115** to face the second carrier substrate **920**.

[0078] The interposer chip **120** may be provided on the second carrier substrate **920**. The interposer chip **120** may be in the embedding region CA of the core portion **110**. The interposer chip **120** may be adhered to the second carrier substrate **920**. In an implementation, the interposer chip **120** may be attached to the second carrier substrate **920** so as to allow the redistribution layer **123** to face the second carrier substrate **920**. Therefore, the redistribution layer **123** of the interposer chip **120** may have a surface at a lower level than that of the bottom surface of the core dielectric pattern **112** of the core portion **110** and at the same level as that of surfaces of the upper core pads **115**.

[0079] Referring to FIG. **15**, an insulating layer **113** may be formed on the second carrier substrate **920**. The insulating layer **113** may fill a gap between the core portion **110** and the interposer chip **120**. In an implementation, an insulating member may be introduced into a gap between the core portion **110** and the interposer chip **120**, and then the insulating member may be cured to form the insulating layer **113**. In an implementation, the insulating member may be formed to cover the interposer chip **120**. Thus, none of the lower chip pads **127** of the interposer chip **120** may be exposed. In addition, a part of the insulating member may cover a top surface of the core portion **110**. In an implementation, the insulating layer **113** may be formed not to cover the top surface of the core portion **110** or a top surface of the interposer chip **120**.

[0080] Referring to FIG. **16**, a third carrier substrate **930** may be attached to the insulating layer **113**. The third carrier substrate **930** may be a dielectric substrate including glass or polymer or a conductive substrate including metal. The third carrier substrate **930** may be provided on its top surface with an adhesive member through which the third carrier substrate **930** may be attached to the insulating layer **113**. In an implementation, the adhesive member may include a glue tape. Thereafter, the second carrier substrate **920** may be removed.

[0081] An upper buildup portion **130** may be formed on the top surfaces of the interposer chip **120** and the core portion **110** that are exposed by the removal of the second carrier substrate **920**. In an implementation, upper dielectric patterns **132** and upper line patterns **134** may be formed on the top surfaces of the core portion **110** and the interposer chip **120**, thereby forming the upper buildup portion **130**. A dielectric layer, such as silicon oxide, may be formed on the top surfaces of the core portion **110** and the interposer chip **120**, and then the dielectric layer may be patterned to form a part of the upper dielectric patterns **132**. The upper dielectric patterns **132** may expose the upper core pads **115** of the core portion **110** and upper chip pads (see **124a**, **124b**, and **124c** of FIG. **2**) of the interposer chip **120**. A conductive layer may be formed on top surface of the upper dielectric patterns **132**, and then the conductive layer may be patterned to form the upper line patterns **134**. The upper line patterns **134** may have corresponding electrical connections with the upper core pads **115** and the upper chip pads (see **124a**, **124b**, and **124c** of FIG. **2**). A dielectric layer may be formed on a top surface of the upper line patterns **134**, and then the dielectric layer may be patterned to form other part of the upper dielectric patterns **132**. The other part of the upper line patterns **134** may be exposed on the upper dielectric patterns **132**.

[0082] Referring to FIG. **17**, a fourth carrier substrate **940** may be attached to the upper buildup

portion **130**. The fourth carrier substrate **940** may be a dielectric substrate including glass or polymer or a conductive substrate including metal. The fourth carrier substrate **940** may be provided on its top surface with an adhesive member through which the fourth carrier substrate **940** may be attached to the upper buildup portion **130**. In an implementation, the adhesive member may include a glue tape. Thereafter, the third carrier substrate **930** may be removed.

[0083] A lower buildup portion **140** may be formed on bottom surfaces of the interposer chip **120** and the core portion **110** that are exposed by the removal of the third carrier substrate **930**. In an implementation, lower dielectric patterns **142** and lower line patterns **144** may be formed on the bottom surfaces of the core portion **110** and the interposer chip **120**, thereby forming the lower buildup portion **140**. The insulating layer **113** may be patterned to expose the lower chip pads **127** of the interposer chip **120** and the lower core pads **117** of the core portion **110**, a conductive layer may be formed on a bottom surface of the insulating layer **113**, and then the conductive layer may be patterned to form a part of the lower line patterns **144**. The lower line patterns **144** may have corresponding electrical connection with the lower chip pads **127** and the lower core pads **117**. A dielectric layer may be formed on bottom surfaces of the lower line patterns **144** and may be patterned to form other parts of the lower dielectric patterns **142**. The lower line patterns **144** may be partially exposed on the lower dielectric patterns **142**.

[0084] Afterward, the fourth carrier substrate **940** may be removed to form a wiring substrate **100**.

[0085] Referring back to FIG. **1**, first and second semiconductor chips **200** and **300** may be mounted on the wiring substrate **100**. The first and second semiconductor chips **200** and **300** may be flip-chip mounted.

[0086] A molding layer **400** may be formed on the wiring substrate **100**. For example, a dielectric material may be coated on the wiring substrate **100** so as to cover the first and second semiconductor chips **200** and **300**, and then the dielectric material may be cured to form the molding layer **400**.

[0087] External terminals **150** may be provided on a bottom surface of the wiring substrate **100**. The external terminals **150** may be formed by providing a soldering member, such as a solder ball or a solder bump, onto surfaces of the lower line patterns **144** that are exposed by patterning the lower dielectric patterns **142**.

[0088] A semiconductor package **1** may be formed through the process mentioned above.

[0089] In an implementation, as illustrated in FIGS. **12** to **17**, a single semiconductor package **1** may be formed on a single wiring substrate **100**. In an implementation, a plurality of semiconductor packages **1** may be formed on a single wiring substrate **100**, and in this case, after the formation of the semiconductor packages **1**, the wiring substrate **100** and the molding layer **400** may undergo a sawing process to separate the semiconductor packages **1** from each other.

[0090] FIGS. **18** to **21** are cross-sectional views of stages in a method of fabricating a semiconductor package according to some example embodiments.

[0091] Referring to FIG. **18**, the interposer chip **120** may be provided on a fifth carrier substrate **950**. The interposer chip **120** may be attached to the fifth carrier substrate **950**. In an implementation, the interposer chip **120** may be attached to the fifth carrier substrate **950** so as to allow the redistribution layer **123** to face the fifth carrier substrate **950**.

[0092] Referring to FIG. **19**, an insulating layer **113-1** may be formed on the fifth carrier substrate **950**. An insulating material may be coated on the fifth carrier substrate **950**, thereby forming the insulating layer **113-1**. The insulating material may be coated to cover the interposer chip **120** on the fifth carrier substrate **950**. The insulating layer **113-1** may cover a lateral surface of the interposer chip **120** and the top surface of the interposer chip **120** (or, a top surface of the chip protection layer **122** and top surfaces of the lower chip pads **127**). The insulating material may include an insulating polymer, such as an epoxy molding compound (EMC).

[0093] Referring to FIG. **20**, through electrodes **114-2** may be formed in the insulating layer **113-1**. In an implementation, via holes VH may be formed to vertically penetrate the insulating layer **113-**

1, and then the via holes VH may be filled with a conductive material to form the through electrodes **114-2**. When the via holes VH are filled with the conductive material, the conductive material may be coated on the insulating layer **113-1** so as to cover a top surface of the insulating layer **113-1**, and the conductive material on the insulating layer **113-1** may be patterned to form a part of the lower line patterns **144** of a lower buildup portion **140** which will be discussed below. [0094] A lower buildup portion **140** may be formed on the insulating layer **113-1**. In an implementation, lower dielectric patterns **142** and lower line patterns **144** may be formed on the top surface of the insulating layer **113-1**, thereby forming the lower buildup portion **140**. The insulating layer **113-1** may be patterned to expose the through electrodes **114-2** and the lower chip pads **127** of the interposer chip **120**, a conductive layer may be formed on the top surface of the insulating layer **113-1**, and then the conductive layer may be patterned to form a part of the lower line patterns **144**. The lower line patterns **144** may have corresponding electrical connections with the lower chip pads **127** and the through electrodes **114-2**. A dielectric layer may be formed on bottom surfaces of the lower line patterns **144** and may be patterned to form other part of the lower dielectric patterns **142**. The lower line patterns **144** may be partially exposed on the lower dielectric patterns **142**.

[0095] Referring to FIG. 21, an upper buildup portion **130** may be formed on surfaces of the insulating layer **113-1** and the interposer chip **120** that are exposed by the removal of the fifth carrier substrate **950**. In an implementation, upper dielectric patterns **132** and upper line patterns **134** may be formed on the surfaces of the insulating layer **113-1** and the interposer chip **120**, thereby forming the upper buildup portion **130**. A dielectric layer, such as silicon oxide, may be formed on the surfaces of the insulating layer **113-1** and the interposer chip **120**, and then the dielectric layer may be patterned to form a part of the upper dielectric patterns **132**. The upper dielectric patterns **132** may expose the through electrodes **114-2** in the insulating layer **113-1** and upper chip pads (see **124a**, **124b**, and **124c** of FIG. 2) of the interposer chip **120**. A conductive layer may be formed on a top surface of the upper dielectric patterns **132**, and then the conductive layer may be patterned to form the upper line patterns **134**. The upper line patterns **134** may have correspond electrical connections with the through electrodes **114-2** and the upper chip pads (see **124a**, **124b**, and **124c** of FIG. 2). A dielectric layer may be formed on top surfaces of the upper line patterns **134**, and then the dielectric layer may be patterned to form other part of the upper dielectric patterns **132**. The upper line patterns **134** may be partially exposed on the upper dielectric patterns **132**. The processes discussed above may form a wiring substrate **100** discussed with reference to FIG. 4.

[0096] Afterwards, identical or similar processes to those discussed in FIG. 1 may be performed. In an implementation, as shown in FIG. 4, first and second semiconductor chips **200** and **300** may be mounted on the wiring substrate **100**. A molding layer **400** may be formed on the wiring substrate **100**, covering the first and second semiconductor chips **200** and **300**. External terminals **150** may be provided on a bottom surface of the wiring substrate **100**.

[0097] The processes mentioned above may form a semiconductor package **3** discussed with reference to FIG. 4.

[0098] In a semiconductor package according to some example embodiments, a high line density may be provided to a redistribution layer of an interposer chip including silicon (Si), and a high degree of freedom of wiring may be given to a wiring substrate. A power or ground signal may be transferred to semiconductor chips through the interposer chip that is individually separated from a core portion that serves as an electrical path along which a drive signal is transferred. Accordingly, it is possible to easily group a power or ground circuit for equalization of the power or ground signal transferred to the semiconductor chips, to effortlessly achieve a wiring design for the drive signal or for the power and ground signals, and to increase the degree of freedom of wiring.

[0099] In addition, high thermal conductive silicon (Si) and the vertically extending vias may facilitate a downward discharge of heat generated from the semiconductor chips mounted on the

wiring substrate. As a result, the semiconductor package may be provided to have high thermal radiation efficiency.

[0100] Moreover, as the semiconductor chips are connected to each other through a redistribution layer having a high line density in the interposer chip, a reduced electrical connection length may be accomplished between the semiconductor chips, and the semiconductor package may have improved electrical characteristics.

[0101] By way of summation and review, in some semiconductor packages, a semiconductor chip may be mounted on a printed circuit board (PCB) and bonding wires or bumps may be used to electrically connect the semiconductor chip to the printed circuit board. With the recent development of the electronics industry, the semiconductor package may be developed with a view toward compact size, small weight, and/or low manufacturing cost. In addition, many kinds of semiconductor packages may show up with the expansion of their application field such as high-capacity mass storage devices.

[0102] One or more embodiments may provide a semiconductor package with a high degree of freedom of wiring.

[0103] One or more embodiments may provide a semiconductor package with increased thermal radiation efficiency.

[0104] One or more embodiments may provide a semiconductor package with improved electrical characteristics.

[0105] Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1.-20. (canceled)

21. A semiconductor package comprising: a substrate including a lower portion, a core portion disposed on the lower portion and an upper portion disposed on the core portion; a first chip disposed in the core portion of the substrate; a second chip disposed on the upper portion of the substrate; a third chip disposed on the upper portion of the substrate; a plurality of external terminals disposed below the lower portion of the substrate; a plurality of upper line patterns disposed on the first chip and including a first line pattern; and a plurality of lower line patterns disposed below the first chip and including a second line pattern, wherein the lower portion of the substrate includes a first dielectric pattern and a third line pattern, wherein the upper portion of the substrate includes a second dielectric pattern and a fourth line pattern, wherein the first chip includes a base layer, a redistribution layer disposed on the base layer, a plurality of vias penetrating the base layer, and a pad disposed below the base layer of the first chip, wherein top surfaces of the plurality of vias are coplanar with a top surface of the base layer, wherein the redistribution layer includes a dielectric layer and a conductive pattern disposed in the dielectric layer, wherein a width of an upper portion of the first line pattern is greater than a width of a lower portion of the first line pattern, and wherein a width of an upper portion of the second line pattern is less than a width of a lower portion of the second line pattern.

22. The semiconductor package of claim 21, wherein the first chip is electrically connected to the lower portion of the substrate and the upper portion of the substrate.

- 23.** The semiconductor package of claim 21, wherein the plurality of vias include a first via that is electrically connected to the redistribution layer and electrically connected to the pad.
- 24.** The semiconductor package of claim 21, wherein the second chip is spaced apart from the third chip.
- 25.** The semiconductor package of claim 21, wherein the second chip is electrically connected to the first chip, and the third chip is electrically connected to the first chip.
- 26.** The semiconductor package of claim 21, wherein the second chip is a logic chip, and the third chip is a memory chip.
- 27.** The semiconductor package of claim 21, further comprising a plurality of through electrodes disposed in the core portion of the substrate.
- 28.** The semiconductor package of claim 21, further comprising: a lower core pad disposed in the core portion of the substrate; an upper core pad disposed in the core portion of the substrate; and a core wiring line contacting the lower core pad and the upper core pad.
- 29.** A semiconductor package comprising: a first wiring layer; an insulating layer disposed on the first wiring layer; a second wiring layer disposed on the insulating layer; a first chip disposed in the insulating layer; a second chip disposed on the second wiring layer; a third chip disposed on the second wiring layer; a conductive element disposed in the insulating layer, disposed between the first wiring layer and the second wiring layer, and spaced apart from the first chip; a molding layer disposed on the second wiring layer, and covering the second chip and the third chip; a plurality of external terminals disposed below the first wiring layer; a first line pattern disposed on the first chip; and a second line pattern disposed below the first chip, wherein the first chip includes a base layer, a redistribution layer disposed on the base layer, and a via penetrating the base layer, wherein a top surface of the via is coplanar with a top surface of the base layer, wherein the redistribution layer includes a dielectric layer and a conductive pattern disposed in the dielectric layer, wherein a width of an upper portion of the first line pattern is greater than a width of a lower portion of the first line pattern, and wherein a width of an upper portion of the second line pattern is less than a width of a lower portion of the second line pattern.
- 30.** The semiconductor package of claim 29, wherein the conductive element includes: a lower core pad disposed on the first wiring layer; an upper core pad disposed in the lower core pad; and a core wiring line contacting the lower core pad and the upper core pad.
- 31.** The semiconductor package of claim 29, wherein the conductive element includes a through electrode.
- 32.** The semiconductor package of claim 31, wherein a width of an upper portion of the through electrode is less than a width of a lower portion of the through electrode.
- 33.** The semiconductor package of claim 31, wherein a width of an upper portion of the through electrode is substantially equal to a width of a lower portion of the through electrode.
- 34.** The semiconductor package of claim 29, wherein the second chip is a logic chip, and the third chip is a memory chip.
- 35.** The semiconductor package of claim 29, wherein the second chip is spaced apart from the third chip.
- 36.** A semiconductor package comprising: a substrate including a lower portion, a core portion disposed on the lower portion and an upper portion disposed on the core portion; a first chip disposed in the core portion of the substrate; a second chip disposed on the upper portion of the substrate; a third chip disposed on the upper portion of the substrate, and spaced apart from the second chip; a first line pattern disposed on the first chip; and a second line pattern disposed below the first chip, wherein the first chip includes a base layer, a redistribution layer disposed on the base layer, and a via penetrating the base layer, wherein a top surface of the via is coplanar with a top surface of the base layer, wherein the redistribution layer includes a dielectric layer and a conductive pattern disposed in the dielectric layer, wherein a width of an upper portion of the first line pattern is greater than a width of a lower portion of the first line pattern, wherein a width of an

- upper portion of the second line pattern is less than a width of a lower portion of the second line pattern, and wherein the second chip is a logic chip, and the third chip is a memory chip.
- 37.** The semiconductor package of claim 36, further comprising: a lower core pad disposed in the core portion of the substrate; an upper core pad disposed in the core portion of the substrate; and a core wiring line contacting the lower core pad and the upper core pad.
- 38.** The semiconductor package of claim 36, further comprising a plurality of through electrodes disposed in the core portion of the substrate.
- 39.** The semiconductor package of claim 36, wherein the first chip includes a pad disposed below the base layer of the first chip.
- 40.** The semiconductor package of claim 36, wherein the lower portion of the substrate includes a first dielectric pattern and a third line pattern, and wherein the upper portion of the substrate includes a second dielectric pattern and a fourth line pattern.
-